Lab Manual - LAB 7 & 08

Semester V – Fall

Group A and B

Course Title: Computer Networks

Bachelor of Science in Software Engineering

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LAB 07 & 08 — Implementing Variable Length Subnet Masking (VLSM) in Cisco Packet Tracer

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Variable Length Subnet Masking (VLSM)

Learning Objectives

By the end of this lab, students will be able to:

- Explain the purpose of Variable Length Subnet Masking (VLSM).
- Distinguish between fixed-length subnetting and VLSM.
- Determine the required subnet size based on host requirements.
- Select appropriate prefixes for different networks.
- Allocate IP addresses using VLSM.
- Configure VLSM-based networks in Cisco Packet Tracer.
- Configure PCs, router interfaces, and default gateways.
- Verify connectivity within and between subnets.
- Identify common VLSM addressing errors.

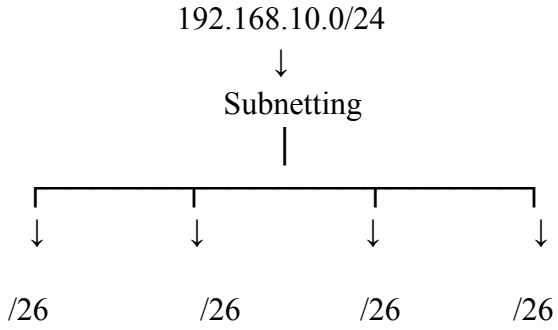
Quick Recap: Subnetting

Since subnetting was covered in the previous lab, we only need a short review.

What is subnetting?

Subnetting is the process of dividing a larger network into smaller networks called **subnets**.

For example:



Each subnet has the **same size** when we divide a /24 into four /26 networks. Each /26 provides:

- 64 total addresses
- 62 usable host addresses

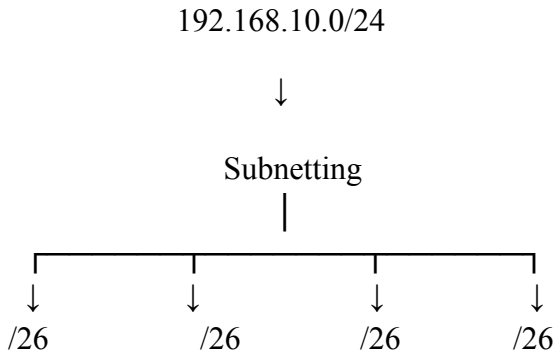
Key terms

Term	Meaning
Subnet mask	Defines the network and host portions of an IP address
Subnet	A smaller network created from a larger network
Subnetting	The process of dividing a larger network into smaller networks

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From Subnetting to VLSM

In traditional fixed-length subnetting, all subnets have the **same size**. For example:



Each subnet provides **62 usable hosts**. But imagine an organization has:

Department	Hosts Required
Data Science	50
Administration	25
Faculty	20
Management	10

Giving every department 62 addresses would waste many addresses. This is where **VLSM** is useful.

What is VLSM?

Variable Length Subnet Masking (VLSM) allows us to create subnets of **different sizes** within the same network. Instead of:

Department	Hosts Required	Prefix
Data Science	50	/26
Administration	25	/26
Faculty	20	/26
Management	10	/26

We can use:

Department	Hosts Required	Prefix
Data Science	50	/26
Administration	25	/27
Faculty	20	/27
Management	10	/28

The prefix is selected according to the number of hosts required.

> **VLSM = Different departments → Different subnet sizes**

Fixed-Length Subnetting vs. VLSM

Fixed-Length Subnetting	VLSM
All subnets have the same size	Subnets can have different sizes
Same prefix for all subnets	Different prefixes can be used

Easier to calculate	Requires more planning
Can waste addresses	Uses addresses more efficiently
Example: /26, /26, /26, /26	Example: /26, /27, /28

Suppose we need:

- Network A → 50 hosts
- Network B → 10 hosts

With fixed-length subnetting:

- A → /26 → 62 hosts
- B → /26 → 62 hosts

With VLSM:

- A → /26 → 62 hosts
- B → /28 → 14 hosts

The second approach uses the available address space more efficiently.

How VLSM Works

VLSM follows a simple process.

Step 1: List the host requirements

Network	Hosts Required
A	50
B	25
C	10
D	5

Step 2: Arrange networks from largest to smallest

Network A: 50
↓
Network B: 25
↓
Network C: 10
↓
Network D: 5

Step 3: Choose the smallest subnet that can accommodate each requirement

Hosts Required	Prefix	Usable Hosts
50	/26	62
25	/27	30
10	/28	14
5	/29	6

Always allocate the largest network first. This prevents address-space fragmentation and makes the allocation easier to manage.

Choosing the Subnet Size

We already know the host formula from the previous lab, where n is the number of host bits.

$$\textbf{Usable hosts} = 2^n - 2$$

For example, if we need **25 hosts**:

- $2^4 - 2 = 14$ Not enough
- $2^5 - 2 = 30$ Enough

Therefore:

- 5 host bits
- 32 total addresses
- /27 prefix
- 30 usable hosts

Common VLSM choices

Required Hosts	Total Addresses Needed	Prefix	Usable Hosts
100	128	/25	126
50	64	/26	62
25	32	/27	30
20	32	/27	30
10	16	/28	14
5	8	/29	6

Activity 01: VLSM Address Allocation

An organization has:

192.168.50.0/24

Determine a suitable prefix and subnet mask for each. Then assign address ranges. It needs:

Department	Hosts Required
Data Science	50
Administration	25
Faculty	20
Management	10

In this case, we need different numbers of hosts, so we use **VLSM (Variable Length Subnet Masking)**. A good approach is to allocate the **largest department first**.

Step 1: Arrange from largest to smallest.

- Data Science → 50
- Administration → 25
- Faculty → 20
- Management → 10

Step 2: Determine the required prefixes

Hosts Required	Minimum Addresses Needed	Prefix	Usable Hosts
50	64	/26	62
25	32	/27	30
20	32	/27	30
10	16	/28	14

Remember:

Usable hosts = Total addresses - 2

Step 3: Allocate the address space

Start with: 192.168.50.0/24

Data Science — 50 hosts

- Needs /26.
- Network: 192.168.50.0/26
- Range: 192.168.50.1 – 192.168.50.62
- Broadcast: 192.168.50.63
- The next available address is 192.168.50.64

Administration — 25 hosts

- Needs /27.
- Network: 192.168.50.64/27
- Range: 192.168.50.65 – 192.168.50.94
- Broadcast: 192.168.50.95

- Next available 192.168.50.96

Faculty — 20 hosts

- Needs /27.
- Network: 192.168.50.96/27
- Range: 192.168.50.97 – 192.168.50.126
- Broadcast: 192.168.50.127
- Next available: 192.168.50.128

Management — 10 hosts

- Needs /28.
- Network: 192.168.50.128/28
- Range: 192.168.50.129 – 192.168.50.142
- Broadcast: 192.168.50.143

Department	Prefix	Mask	Network	Host Range	Broadcast
Data Science	/26	255.255.255.192	192.168.50.0	192.168.50.1 – 192.168.50.62	192.168.50.63
Administration	/27	255.255.255.224	192.168.50.64	192.168.50.65 – 192.168.50.94	192.168.50.95
Faculty	/27	255.255.255.224	192.168.50.96	192.168.50.97 – 192.168.50.126	192.168.50.127
Management	/28	255.255.255.240	192.168.50.128	192.168.50.129 – 192.168.50.142	192.168.50.143

The given addresses are still available for future use.

192.168.50.144 – 192.168.50.255

This is **Variable Length Subnet Masking**.

Activity 02: Practice VLSM Calculation

You are given:

192.168.100.0/24

The organization requires:

Network	Hosts Required
Network A	60
Network B	30
Network C	12
Network D	5

Tasks

1. Arrange the networks from largest to smallest.
2. Determine the required prefix for each network.
3. Determine the subnet mask.
4. Allocate the network addresses.
5. Find:
    - First usable host
    - Last usable host
    - Broadcast address

Network	Prefix	Subnet Mask	Network	Host Range	Broadcast
A	/26	255.255.255.192	192.168.100.0	192.168.100.1 – 192.168.100.62	192.168.50.63
B	/27	255.255.255.224	192.168.100.64	192.168.100.65 – 192.168.100.94	192.168.50.95
C	/28	255.255.255.240	192.168.100.96	192.168.100.97 – 192.168.100.109	192.168.50.110
D	/29	255.255.255.248	192.168.100.111	192.168.110.112 – 192.168.100.117	192.168.50.118

6. Identify the remaining unused address space.

192.168.100.119 – 192.168.100.255

Activity 03: Build the VLSM Network

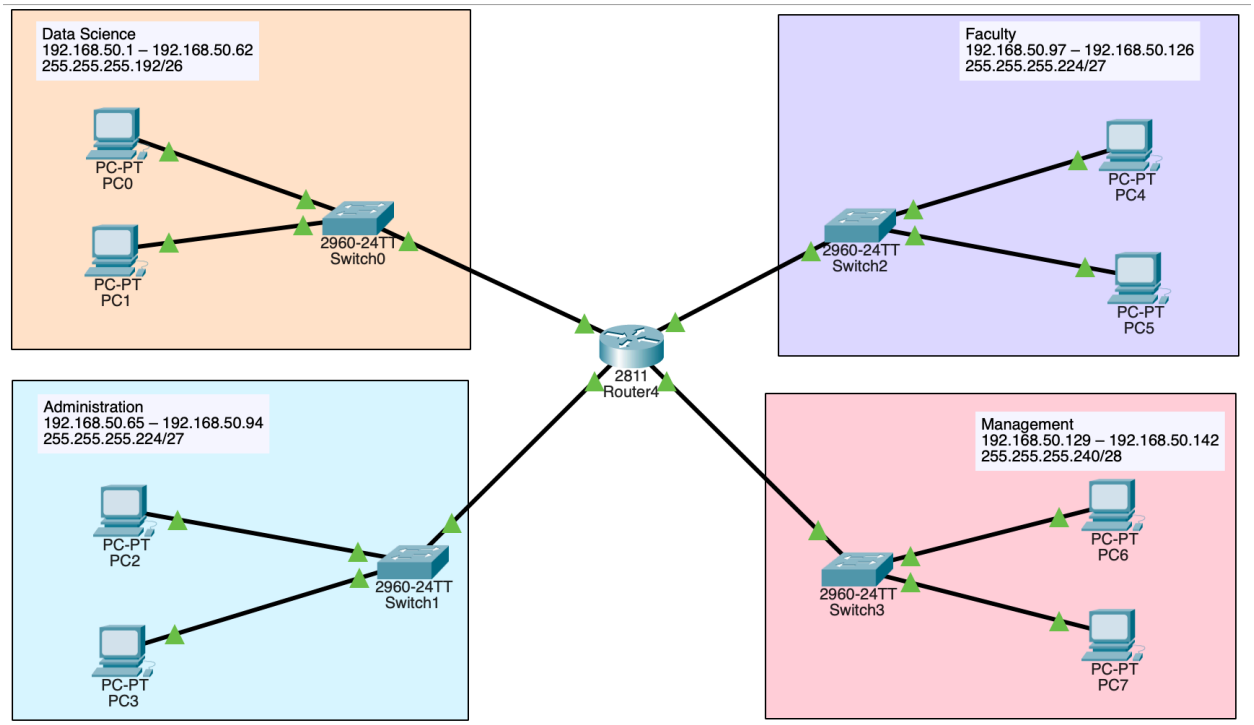
Use the VLSM design from Activity 01 with the given IP 192.168.50.0/24

Department	Prefix	Mask	Network	Host Range	Broadcast
Data Science	/26	255.255.255.192	192.168.50.0	192.168.50.1 – 192.168.50.62	192.168.50.63
Administration	/27	255.255.255.224	192.168.50.64	192.168.50.65 – 192.168.50.94	192.168.50.95
Faculty	/27	255.255.255.224	192.168.50.96	192.168.50.97 – 192.168.50.126	192.168.50.127
Management	/28	255.255.255.240	192.168.50.128	192.168.50.129 – 192.168.50.142	192.168.50.143

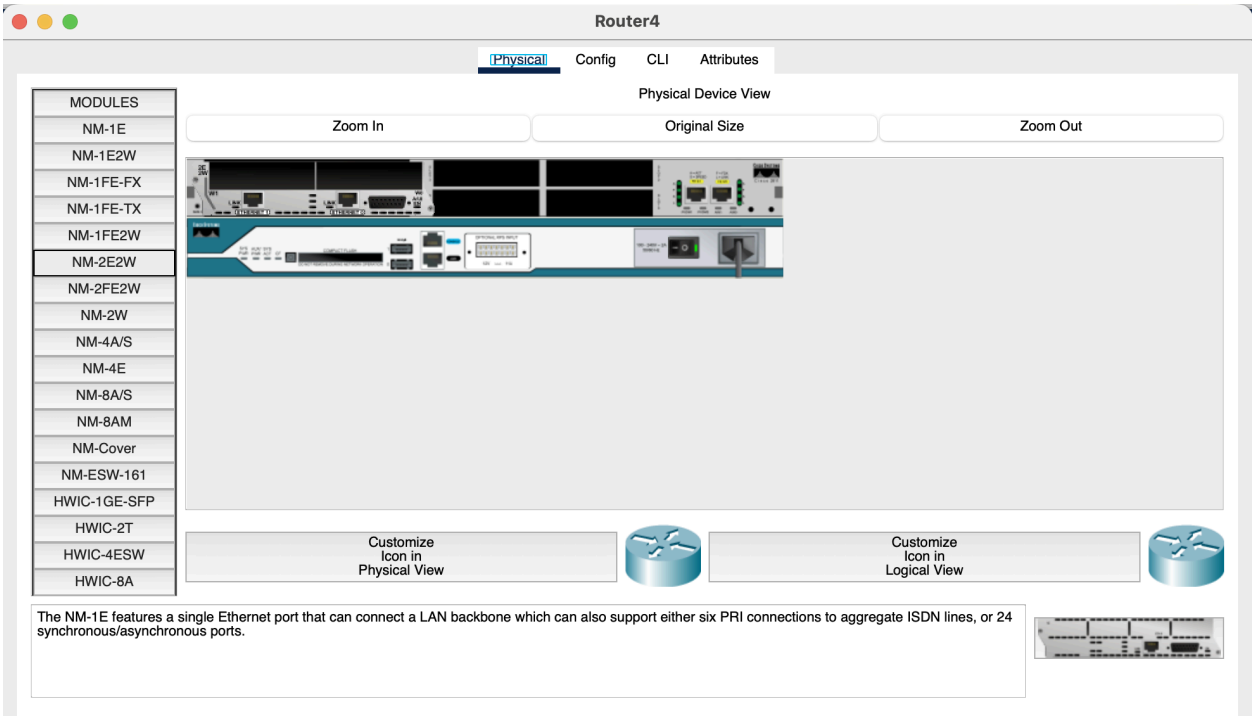
The given addresses are still available for future use.

192.168.50.144 – 192.168.50.255

Topology



- Add Module NM-2E2W to Router to add 2 more Ethernet Ports for connection.



Assign IP addresses from the correct subnet.

Department	PC	IP Address	Subnet Mask	Default Gateway
Data Science	PC0	192.168.50.2	255.255.255.192	192.168.50.1
	PC1	192.168.50.3	255.255.255.192	192.168.50.1
Administration	PC2	192.168.50.66	255.255.255.224	192.168.50.65
	PC3	192.168.50.67	255.255.255.224	192.168.50.65
Faculty	PC4	192.168.50.98	255.255.255.224	192.168.50.97
	PC5	192.168.50.99	255.255.255.224	192.168.50.97
Management	PC6	192.168.50.130	255.255.255.240	192.168.50.129
	PC7	192.168.50.131	255.255.255.240	192.168.50.129

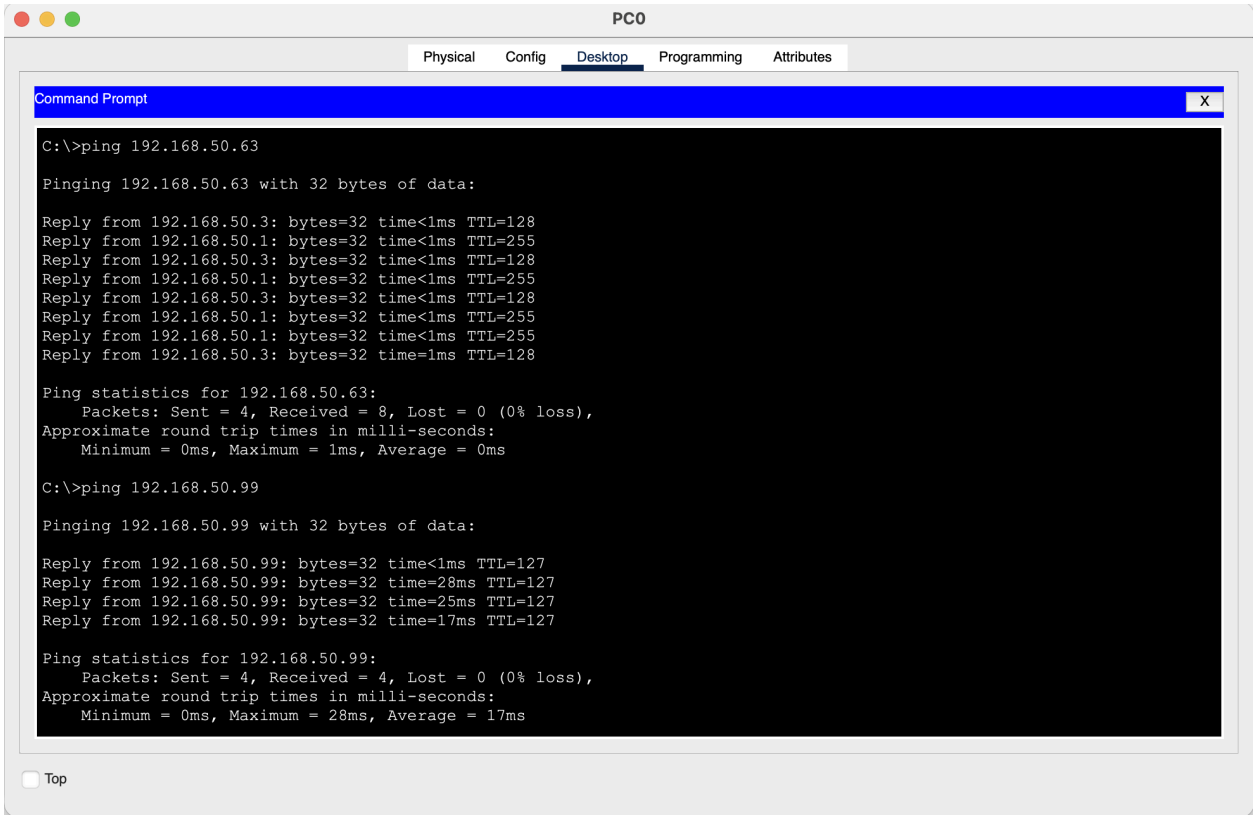
Router gateway addresses

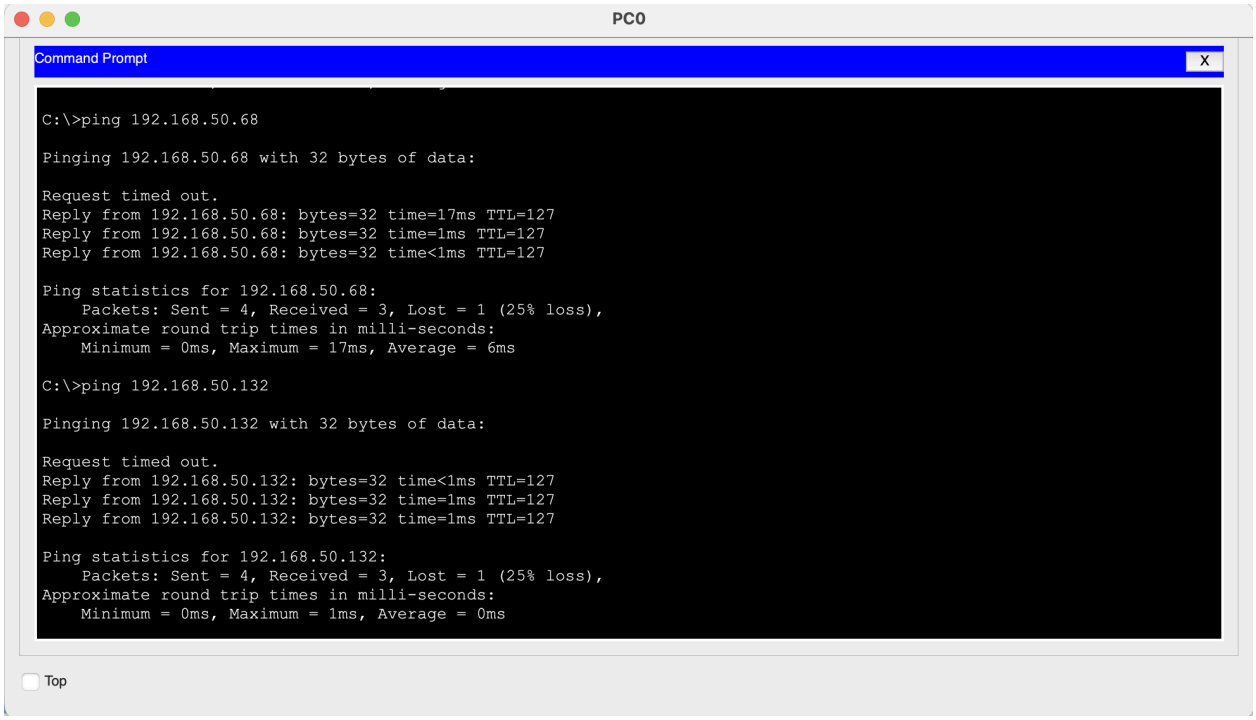
For your Packet Tracer configuration, the router interfaces would use:

Department	Router Interface IP	Subnet Mask
Data Science	192.168.50.1	255.255.255.192
Administration	192.168.50.65	255.255.255.224
Faculty	192.168.50.97	255.255.255.224
Management	192.168.50.129	255.255.255.240

Test Connectivity

- Broadcast in each network
- Ping from PC 0 to any PC from each network.





Common VLSM Errors

Error 1: Choosing a subnet that is too small

50 hosts → /27

- Wrong because /27 provides only 30 usable hosts.
- Correct: 50 hosts → /26

Error 2: Forgetting the network and broadcast addresses

- A /28 has 16 total addresses. 14 usable hosts, not 16 usable hosts.

Error 3: Allocating networks in random order

- Always start with the **largest requirement**.

Error 4: Overlapping subnets

For example:

- Network A → 192.168.50.0/26
- Network B → 192.168.50.32/27

These overlap. Make sure that each subnet begins after the previous subnet's address space ends.

Error 5: Wrong default gateway

The gateway should normally be a **usable host address within that subnet**, commonly the first usable address in the lab design.
